

Lindblad as a Physical Oracle Layer: Hardware-Anchored Attestation for Real-World Data

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Abstract

We extend the Lindblad Cryptography Protocol (LCP) — previously applied to consensus and decentralized finance — to the problem of verifying real-world data on-chain. Existing oracle protocols solve the immutability of records on-chain but inherit a structural weakness at the data ingestion layer: the data still originates in software, run by a trusted operator, and can be fabricated at the source before being recorded. We show that hardware with silicon-derived unforgeable identity (SRAM PUF + BCH fuzzy extractor) can sign measurements directly, producing attestations that are cryptographically verifiable by any third party without trust in the operator. We demonstrate end-to-end validation on mainnet using a live commodity price (West Texas Intermediate crude oil) sourced from the U.S. Energy Information Administration, signed by a physical node, and verified by a publicly accessible mathematical check. We further describe the generalization of this primitive across five application verticals: agriculture, energy, mining and resource extraction, Real-World Asset (RWA) tokenization, and verified ad delivery. The Lindblad Oracle complements existing oracle protocols (Chainlink, API3, UMA) by providing a hardware-anchored root of trust at the data-origination layer, beneath their data-distribution layer.

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1 Introduction

Smart contracts can execute on-chain logic with mathematical certainty, but most useful applications require information about the world outside the chain: prices, weather, sensor readings, asset locations, ad impressions, energy generation. Bridging this gap is the well-known *oracle problem* [5]: how does a deterministic on-chain system reliably know about non-deterministic real-world events?

The dominant solution today — exemplified by Chainlink, API3, UMA, and similar protocols — solves this by aggregating reports from multiple off-chain operators, applying consensus or staking-based incentives, and publishing the result on-chain in a tamper-resistant way [4]. This works well for the *transport* of data from off-chain sources to on-chain consumers. It does not address the question of where the data originated and whether *that origin* can be trusted.

In practice, the input to most oracle networks is itself produced by software: an API endpoint, a database query, a web scraper. The oracle’s job is to faithfully relay that input, not to verify it. If the data is manipulated before reaching the oracle — by the operator of the source API, by a man-in-the-middle, or by a compromised sensor reporting fake values — the on-chain record is still tamper-resistant, but it records a lie.

This paper takes a different approach: anchor the data itself in physical hardware whose identity cannot be forged, before any transport layer is involved. We extend the Lindblad Cryptography Protocol (LCP), previously applied to consensus [1] and DeFi [2], to act as a *physical oracle layer*: a base layer where any piece of data, before being relayed anywhere, is signed by a specific physical device with a unique silicon-derived identity, timestamped by physically-generated entropy, and recorded permanently and verifiably.

Chainlink, API3, and similar protocols operate at the data-distribution layer; the Lindblad Oracle operates one layer below, at the data-origination layer, providing a root-of-trust at the source that existing oracle networks could themselves consume.

2 The Oracle Problem Reconsidered

2.1 Where Existing Oracles Are Strong

Modern oracle networks excel at the following:

- **Sybil resistance via staking** — operators with bonded capital are economically disincentivized from misreporting.
- **Aggregation across reporters** — by requiring agreement among multiple independent operators, individual operator failures do not corrupt the aggregate.
- **On-chain immutability** — once published, data records cannot be altered without consensus-level attacks against the underlying blockchain.

These properties are valuable. Many DeFi protocols would not function without them.

2.2 Where the Trust Gap Remains

What these protocols do not address — and explicitly are not designed to address — is the integrity of the data at its point of origin. An oracle network reporting that “WTI crude oil traded at \$95/bbl at time T ” can prove that multiple operators agreed on this value and that the record has not been altered since. It cannot prove that any of those operators actually observed the value at the source, that the source was not manipulated, or that a compromised reporter did not substitute a different value.

Aggregation-based oracles are deliberately agnostic to data provenance — by design. The implicit assumption is that the source is honest, or that aggregation across many sources will average out individual dishonesty. For most use cases, this is a reasonable assumption. For high-value or adversarial use cases — large insurance payouts, commodity provenance with regulatory consequences, asset-backed lending with significant collateral — this assumption is increasingly questioned.

Recent industry attention to ad-delivery verification [6], regulatory pressure on critical-mineral supply chains [7], and the persistent failure of RWA tokenization protocols to maintain trust in off-chain collateral [8] all point to the same underlying gap: the data itself, at the point where it enters the digital world, has no cryptographic root.

3 The Lindblad Cryptography Protocol as Background

The Lindblad Oracle applies the existing LCP stack [1] to a new use case rather than introducing new cryptographic constructions. We summarize the four layers briefly, since each plays a role in the oracle architecture:

- **L1 — SRAM PUF.** A Physically Unclonable Function derives a per-chip fingerprint from manufacturing-time variance in SRAM cell power-up states. Stabilized via a BCH(255,139,t=15) fuzzy extractor with validated hardware metrics: 86% rock-stable bits, 0.00% intra-device Hamming distance over 12-cycle enrollment, 48.6% inter-device distance.
- **L2 — P-256 ECDSA.** The fuzzy-extracted key seeds a NIST P-256 (secp256r1) keypair, fully compatible with standard cryptographic libraries (Python `cryptography`, JavaScript `noble-curves`, Go `crypto/ecdsa`, OpenSSL). The private key is regenerated from the PUF at each boot and never persists to storage.
- **L3 — Hybrid Stochastic Chua (HSC).** A chaotic oscillator driven by thermal ADC noise produces nonces that are physically unrepeatable, providing time-stamping that cannot be predicted or forged even by the same hardware.
- **L4 — Lindblad Consensus.** A dissipative LoRa mesh network propagates state updates and converges to a Lindblad-master-equation steady state, providing distributed agreement without proof-of-work or proof-of-stake.

For the oracle application, layers L1–L3 are the essential primitives: identity, signing, and unforgeable timestamping. Layer L4 propagates the resulting attestations through the network.

4 The Physical Attestation Pattern

4.1 Definition

A **physical attestation** is a tuple

$$\alpha = \langle d, t, c_t, h, \sigma, \text{pk} \rangle \quad (1)$$

where:

- d is the data being attested (e.g., a measured value, a price quote, a sensor reading)
- t is the timestamp at which d was observed
- c_t is the Chua HSC nonce generated at the moment of signing (L3)
- $h = \text{SHA-256}(\text{concat}(d, \text{nodeId}, t, c_t))$ is the message digest

- $\sigma = (r, s)$ is the ECDSA-P256 signature of h under the node’s private key, where the private key is derived from the SRAM PUF (L1+L2)
- pk is the corresponding public key, registered publicly

Verification consists of:

1. Reconstructing h from the public components $(d, \text{nodeId}, t, c_t)$.
2. Verifying σ against h using pk via standard ECDSA-P256.

A failed verification implies that at least one of the following must be true: d , nodeId , t , or c_t has been modified; σ was generated by a different private key; or the entire tuple is fabricated.

4.2 Properties

Source authenticity. Because pk is the public key corresponding to a private key that exists only inside the silicon of a specific chip, a valid signature implies that *that specific physical chip* signed h . No software process running anywhere else can produce a valid signature.

Temporal authenticity. The Chua HSC nonce c_t is generated from thermal noise sampled at signing time. It cannot be precomputed and is essentially unrepeatable; even the same hardware producing two attestations of the same data d will produce different c_t values, yielding different signatures.

External verifiability. Verification requires only public information: $d, t, c_t, r, s, \text{nodeId}$, and the publicly-registered pk . No proprietary code, no protocol-specific library, and no trust in the operator of the registry is required. Any participant with a standard ECDSA library can verify.

Layered security (Kerckhoffs’s principle). The security of the attestation does not depend on any aspect of the protocol being secret. The message construction is public, the algorithm is public (NIST-standard P-256), the public key is public. Only the private key — which exists physically and only inside the chip — is secret.

5 Implementation and Live Validation

5.1 Architecture

The Lindblad Oracle is implemented as an extension to the Lindblad fullnode, exposing the following public HTTP endpoints:

- `POST /attestations/commodity/quote` — initiates an attestation by querying an external data source and creating a pending signature request for a designated node
- `POST /attestations/commodity/submit` — finalizes an attestation once the physical node has signed the challenge
- `GET /attestations?category=...` — queries recorded attestations
- `GET /node-keys` — lists registered public keys for all physical nodes
- `GET /node-keys/{nodeId}` — retrieves the public key for a specific node
- `GET /verify-12/{signId}` — performs full ECDSA verification of a signature against the node’s stored public key, returning a binary result with the reconstructed message digest
- `GET /get-sign?signId={id}` — returns the raw signature components needed for offline verification

5.2 End-to-End Validation: WTI Crude Oil

The following sequence was executed on mainnet on June 12, 2026, using physical node LD95A821D:

1. The fullnode queried the U.S. Energy Information Administration (EIA) public API for the most recent spot price of West Texas Intermediate crude oil (series RWTC).
2. The fullnode constructed a challenge `commodity_RWTC_95_2026-06-08` and created a pending signature request with identifier `0e8029bb`.
3. Node LD95A821D detected the pending request via its polling loop, generated a Chua HSC nonce (`AF6944A5`), constructed the message

```
msg = "commodity_RWTC_95_2026-06-08" + "LD95A821D"
      + "1781307465" + "AF6944A5"
```

and computed

```
h = SHA-256(msg) = dd662f6727605b1c...
```

4. The node signed `h` using its PUF-derived private key, producing

```
r = 4d46397890b5c3d8...
s = 8df4008085730806...
```

and submitted `(r, s, puf, chuaNonce, ts)` to the fullnode.

5. The fullnode recorded the attestation as `05def9656afa`.

The node's public key, registered in the `node_keys` table during initial onboarding, is:

```
0429da61e9a4562b28b3dbabf33ad00de6a09716e81a3925aa81024988b9008b
c137e511e7e8037d149f377fbae81a4531037b1c6a8f0d8ed378488a415348ffe5
```

This is uncompressed P-256 format: the leading `04` prefix indicates uncompressed encoding, followed by 32 bytes of X coordinate and 32 bytes of Y coordinate.

5.3 Independent Verification in Python

The attestation can be verified by any party using only standard tools. The following Python script, runnable from any machine with internet access:

```
import hashlib, requests
from cryptography.hazmat.primitives.asymmetric import ec
from cryptography.hazmat.primitives.asymmetric.utils \
    import encode_dss_signature
from cryptography.hazmat.primitives import hashes

SIGN_ID = "0e8029bb"
NODE_ID = "LD95A821D"

# 1. Fetch signature data (public endpoint)
sign = requests.get(
    f"https://lindblad.io/get-sign?signId={SIGN_ID}").json()

# 2. Fetch the node's public key (public endpoint)
key = requests.get(
    f"https://lindblad.io/node-keys/{NODE_ID}").json()
```

```

# 3. Reconstruct the exact message the firmware signed
msg = (sign["challenge"] + NODE_ID
      + str(sign["ts"]) + sign["chuaNonce"].upper())

# 4. Verify with standard ECDSA-P256
pub = ec.EllipticCurvePublicKey.from_encoded_point(
    ec.SECP256R1(), bytes.fromhex(key["public_key"]))
sig = encode_dss_signature(
    int(sign["r"], 16), int(sign["s"], 16))

pub.verify(sig, msg.encode(), ec.ECDSA(hashes.SHA256()))
print("Signature valid")

```

Execution of this script against the live `lindblad.io` endpoints produces a successful verification, confirming externally that the attestation was generated by node `LD95A821D`'s private key.

5.4 Coverage Across Operations

The same verification primitive is applied to all PUF-signed operations in the Lindblad ecosystem, not only oracle attestations. As of writing, four operation classes are covered:

- LindFi liquidity additions (`/pool/add`)
- LindFi liquidity removals (`/pool/remove`)
- LindFi swaps (`/pool/swap`)
- Oracle commodity attestations (`/attestations/commodity/submit`)

Each operation response includes the fields `level2_verified` (boolean) and `level2_reason` (string), providing the verification result alongside the operation result. The system operates in observer mode at the time of writing: verification is computed but not used to reject operations, allowing accumulation of empirical data before enforcement.

6 Application Verticals

The oracle primitive described above is domain-agnostic. The data being attested can be a financial quantity, a sensor reading, a status indicator, or any value measurable in the physical world. We describe five verticals in which this primitive addresses a real, paid market.

6.1 Agriculture

Soil moisture, temperature, and weather sensors equipped with LCP firmware can sign their measurements directly, enabling parametric crop insurance with automated payouts (no adjuster required), origin certification with continuous evidence (no batch testing), and verifiable carbon-sequestration credits backed by continuous physical measurement.

Today, the global crop insurance market relies on remote-sensing data and manual loss adjustment. Both layers introduce significant cost and dispute risk. Physical attestation eliminates the verification burden at its source.

6.2 Energy

Smart meters, solar inverters, and battery management systems with LCP firmware can sign their own generation, consumption, or storage events. This enables Renewable Energy Credits (RECs) issued directly from the device (rather than via certifying intermediaries), peer-to-peer

energy trading with verifiable settlement, and grid demand-response programs where payments depend on cryptographically attested actions rather than self-reported metering.

The current REC market depends on a chain of trusted intermediaries between generation and certification. LCP collapses this chain to a single verifiable signature.

6.3 Mining and Resource Extraction

Critical-mineral supply chains face increasing regulatory pressure to demonstrate provenance — particularly for lithium, cobalt, rare earths, and conflict-sensitive materials. LCP-equipped weighing systems, conveyor sensors, and extraction telemetry can sign volume and location data at the point of extraction, providing an unforgeable chain of custody from mine to refinery.

For nations with significant resource reserves and emerging downstream processing capacity, sovereign attestation infrastructure based on LCP would provide independent verification of extraction figures without dependence on external certification firms.

6.4 Real-World Asset (RWA) Tokenization

Tokenized real-world assets — invoices, commodities, real estate, equipment — typically face a structural problem: the asset is verified once at minting and never again. If the asset is sold, damaged, or never existed, the on-chain token continues to circulate. LCP-equipped condition and location sensors attached to or embedded in physical assets provide *continuous re-verification*: the token’s value is dynamically backed by an unbroken stream of attestations from the underlying asset itself.

6.5 Verified Ad Delivery

The recent LG Electronics / Arbitrum pilot [6] demonstrates industry recognition of the ad-fraud problem and the use of on-chain immutability to address record-keeping. However, the initial record still originates in software (an SDK or app) running on the device. LCP-equipped displays would sign delivery events at the silicon level, addressing the fraud problem at its root: a bot or simulated impression has no corresponding physical device with a valid PUF signature.

7 Relationship to Existing Oracle Protocols

The Lindblad Oracle complements rather than replaces Chainlink, API3, UMA, and similar protocols. Those operate at the data-distribution layer: they take an input value and reliably deliver it to a consumer smart contract. The Lindblad Oracle operates at the data-origination layer: it produces input values that carry cryptographic provenance.

A natural composition emerges: an LCP-attested measurement can be consumed by a Chainlink node, aggregated with other sources, and delivered to a smart contract. The Chainlink layer continues to provide what it does well — Sybil-resistant aggregation and on-chain delivery — while the LCP layer adds source authenticity that aggregation alone cannot provide.

Table 1: Layered Trust Model: Origination vs. Distribution

Layer	Concern	Example Protocol
Data origination	Source authenticity	Lindblad Oracle (this work)
Data distribution	Aggregation, delivery	Chainlink, API3, UMA
On-chain consumption	Smart contract logic	EVM, Solana, etc.

8 Limitations and Future Work

We note several honest limitations of the work presented:

- **Hardware availability.** LCP requires firmware running on devices with SRAM exhibiting PUF behavior. While this is widespread (ESP32, many MCUs), it is not universal. Pure-software systems cannot generate LCP-compliant attestations.
- **Sensor compromise.** The protocol guarantees that a specific physical chip signed a specific value at a specific time. It does not guarantee that the value the chip measured was honest. A sensor whose input is tampered with (e.g., a thermometer placed near a heat source) will faithfully sign incorrect data. Physical security of the sensor environment remains the operator’s responsibility.
- **Public-key registration trust.** A node’s public key must be registered with the protocol once, typically at onboarding. The integrity of this initial registration depends on out-of-band trust (the operator confirms that public key pk corresponds to physical chip with PUF p). Subsequent verifications inherit this initial trust assumption.
- **Sample size for live validation.** As of writing, three physical nodes have produced PUF-signed operations on mainnet. While this is sufficient to demonstrate the protocol, statistical analyses of PCV-4 components for distinguishing genuine from compromised hardware will become more robust as additional nodes accumulate signature history.

Future work includes: enforcement mode for Level-2 verification (currently observer mode); automated public-key onboarding; standardized attestation schemas per vertical; firmware support for specialized sensors beyond the current generic Heltec/ESP32 platform; and integration with downstream oracle protocols for combined origination + distribution flows.

9 Conclusion

The oracle problem is conventionally treated as a problem of consensus among reporters delivering data to a blockchain. We argue that a complementary primitive exists below this layer: the origination of the data itself in physical hardware with unforgeable identity. The Lindblad Cryptography Protocol provides this primitive, with the same architecture previously demonstrated for consensus and DeFi applications, now extended to a public attestation registry.

We have demonstrated end-to-end validation on mainnet, with externally verifiable signatures of real-world commodity data. The infrastructure required to verify a Lindblad attestation consists of two public HTTP endpoints and a standard ECDSA library; no proprietary code or trusted intermediary is required. The same primitive generalizes across five identified application verticals, all of which currently depend on intermediary certification that the Lindblad Oracle could replace or augment.

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Lindblad Protocol — The hardware decides. The physics guarantees. The chain records.